

Operating Systems 305

Group Project

Task 2

Part I: An xv6 Lottery Scheduler

You are required to implement the project details specified in:

<https://github.com/remzi-arpacidusseau/ostep-projects/tree/master/scheduling-xv6-lottery>

To check that your implementation is correct: run the tests provided with the project.

For more information on how the (more complex) lottery scheduler works: see

<https://pages.cs.wisc.edu/~remzi/OSTEP/cpu-sched-lottery.pdf>

Part II: Answer the following:

Assume that tickets are not chosen at random, i.e., the scheduler is unfair. What would be required to prevent starvation? Explain how you would implement your solution and show that it prevents starvation. [You are not required to code the solution]